

Almost All Real Numbers Are Normal

A Comprehensive Report on Proof of
Almost All Real Numbers Are Normal
via Probability and Combinatorics

Abstract

In 1909, Émile Borel introduced the concept of *normal numbers* and proved that almost all real numbers are absolutely normal. This report presents a comprehensive treatment of Borel's theorem from three distinct perspectives. We begin with foundational definitions and the Lebesgue measure, then present Borel's original probabilistic proof. Next, we reproduce Felix Hausdorff's 1914 combinatorial argument for base 2 and its generalisation to arbitrary bases following Edmund Weitz.

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1 Definitions and Preliminaries

1.1 Normal Numbers

Definition 1.1 (Simply normal). For $b \in \{0, \dots, r-1\}$, define

$$p_{b,n,\alpha,r} = \left| \{j \in \mathbb{N}^+ : j \leq n \wedge \alpha_{j,r} = b\} \right|,$$

which counts how often the digit b occurs among the first n digits of α after the radix point. A number $\alpha \in \mathbb{R}$ is called **simply normal in base r** if

$$\lim_{n \rightarrow \infty} \frac{p_{b,n,\alpha,r}}{n} = \frac{1}{r} \quad (1)$$

holds for every digit $b \in \Sigma_r$.

Definition 1.2 (Normal and absolutely normal). The number α is called **normal in base r** if $r^m \alpha$ is simply normal in base r^n for all $n \in \mathbb{N}^+$ and all $m \in \mathbb{N}$. The number α is called **absolutely normal** if it is normal in every integer base $r \geq 2$.

Note: Measures and the Lebesgue Measure

A **measure** is a function μ that assigns a non-negative extended real number (i.e. a value in $[0, \infty]$) to certain subsets of a given set X , satisfying two axioms:

- (1) $\mu(\emptyset) = 0$.
- (2) **σ -additivity**: if $A_1, A_2, \dots$ are pairwise disjoint measurable sets, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

The pair $(X, \mathcal{A})$ where $\mathcal{A}$ is a σ -algebra of subsets of X , together with a measure μ , is called a *measure space* $(X, \mathcal{A}, \mu)$.

The Lebesgue measure. The most important measure on $\mathbb{R}$ is the **Lebesgue measure**, denoted λ (or sometimes μ). It is the unique measure on the Borel σ -algebra of $\mathbb{R}$ satisfying:

- $\lambda([a, b]) = b - a$ for every interval $[a, b]$ (and likewise for open, half-open intervals).
- λ is **translation-invariant**: $\lambda(A + c) = \lambda(A)$ for all measurable A and $c \in \mathbb{R}$.

It follows that finite sets have measure zero, since each singleton $\{x\}$ is an interval of length 0.

Null sets. A set A is called a **null set** (or a set of *measure zero*) if $\lambda(A) = 0$. Equivalently, A is a null set if, for every $\varepsilon > 0$, one can find countably many intervals whose union covers A and whose total length is at most ε .

“Almost all.” If A is a set with positive measure and P is a property elements of A can have or not, we say **almost all** elements of A have property P if the set of elements *not* having P is a null set.

1.2 An Example of a Normal Number

Example 1.3 (The Champernowne number). The **Champernowne number** (in base 10) is obtained by concatenating all positive integers in order after the decimal point:

$$C_{10} = 0.1\,2\,3\,4\,5\,6\,7\,8\,9\,10\,11\,12\,13\,14\,15\,16\,\dots$$

This number is normal in base 10.

Why is C_{10} normal? The observation is that the k -digit positive integers run from 10^{k-1} to $10^k - 1$, and there are $9 \cdot 10^{k-1}$ of them. Among all k -digit integers, every digit $d \in \{0, \dots, 9\}$ appears in each of the k positions with roughly the same frequency (with a minor correction for leading digits). As $n \rightarrow \infty$, the contribution from k -digit numbers dominates for ever-larger k , and a counting argument shows that the limiting frequency of every single digit is $1/10$, every block of two digits has frequency $1/100$, and so on. Hence C_{10} is normal in base 10. A full proof of this fact was given by Champernowne himself in 1933 [2].

Note, however, that proving C_{10} is *absolutely* normal (i.e. normal in *every* base) remains an open problem.

2 Borel’s Proof: Almost All Numbers in $[0, 1)$ Are Absolutely Normal

We now present Borel’s original 1909 argument, which is probabilistic in nature. The digits of the base- q expansion of a “random” number in $[0, 1)$ behave like independent, identically distributed random variables, each uniform on $\{0, 1, \dots, q-1\}$.

Theorem 2.1. *Almost all real numbers in the interval $[0, 1)$ are absolutely normal.*

Proof. Let μ denote the Lebesgue measure on $[0, 1)$. Fix an integer base $q \geq 2$. Any $x \in [0, 1)$ has a unique base- q expansion (outside a countable, hence measure-zero, set of terminating fractions):

$$x = \sum_{i=1}^{\infty} \frac{x_i}{q^i}, \quad x_i \in \{0, 1, \dots, q-1\}.$$

Fix a digit $d \in \{0, \dots, q-1\}$ and let $S_n(x)$ count the number of times d appears among the first n digits. Set $p = 1/q$ and $p' = 1 - 1/q$.

Let $k = np + s$, where s is the deviation from the expected count. The probability of exactly k occurrences is

$$P(S_n = k) = \binom{n}{k} p^k (p')^{n-k}.$$

Applying Stirling's approximation $N! \sim N^N e^{-N} \sqrt{2\pi N}$ and the Taylor expansion $\ln(1 + u) \approx u - u^2/2$ for small u , the linear terms cancel and we obtain

$$\ln P(np + s) \approx -\frac{s^2}{2npp'}.$$

Exponentiating gives the De Moivre-Laplace bound:

$$P(np + s) \approx \frac{1}{\sqrt{2\pi npp'}} e^{-s^2/(2npp')}.$$

Now, choose $\varepsilon \in (0, 1/2)$ and set $s = n^{1/2+\varepsilon}$. Let E_n be the set of $x \in [0, 1)$ with $|S_n(x) - np| \geq n^{1/2+\varepsilon}$. Integrating the Gaussian tails:

$$\mu(E_n) \leq A e^{-cn^{2\varepsilon}}$$

for positive constants A, c depending on p and p' .

Because $e^{-cn^{2\varepsilon}}$ decays faster than any inverse polynomial:

$$\sum_{n=1}^{\infty} \mu(E_n) < \infty.$$

Let $E_\infty = \limsup_{n \rightarrow \infty} E_n$. That is now called the **Borel-Cantelli lemma**, $\mu(E_\infty) = 0$. Hence for almost every $x \in [0, 1)$, there exists N_x such that for all $n \geq N_x$:

$$|S_n(x) - np| < n^{1/2+\varepsilon}.$$

Dividing by n :

$$\left| \frac{S_n(x)}{n} - p \right| < \frac{1}{n^{1/2-\varepsilon}} \xrightarrow{n \rightarrow \infty} 0.$$

Thus $\lim_{n \rightarrow \infty} S_n(x)/n = 1/q$ for almost all x . Since the set of digits is finite, the intersection of these q sets of full measure is still of full measure. Hence almost all $x \in [0, 1)$ are simply normal in base q .

Digits grouped into blocks of length k in base q are equivalent to single digits in base $Q = q^k$. Applying Steps 1–3 with base Q , almost all x are simply normal in base q^k . Since k ranges over the countable set $\mathbb{N}^+$, the intersection of countably many sets of full measure is still of full measure. Hence almost all $x \in [0, 1)$ are normal in base q .

The set of bases $\{2, 3, 4, \dots\}$ is countable. The union of countably many measure-zero sets (the “non-normal in base q ” sets) is measure zero. Hence almost all $x \in [0, 1)$ are absolutely normal. $\square$

Note: The Modern Approach via the Strong Law of Large Numbers (SLLN)

In modern probability, Steps 1–3 are replaced by a single invocation of the **Strong Law of Large Numbers**. By viewing $([0, 1), \mathcal{B}, \mu)$ as a probability space, and the digits x_i as i.i.d. random variables. For the indicator $I_i(x) = \mathbf{1}_{[x_i=d]}$, we have $\mathbb{E}[I_i] = 1/q$ and $S_n(x) = \sum_{i=1}^n I_i(x)$. The SLLN gives

$$\lim_{n \rightarrow \infty} \frac{S_n(x)}{n} = \frac{1}{q} \quad \text{for almost all } x \in [0, 1).$$

However, Borel published his proof in 1909, *before* the SLLN existed. By manually bounding deviations via Stirling's formula and introducing denumerable intersections, Borel effectively invented and proved the SLLN for Bernoulli trials.

3 Hausdorff's Proof for Base 2 (Without i.i.d. Assumption)

In 1914, Felix Hausdorff [3] published a proof that almost all numbers are simply normal in base 2. His argument is purely combinatorial and measure-theoretic, it does *not* invoke the the assumption that digits are i.i.d. random variables. We now reproduce his proof for base 2.

3.1 Sets of Dyadic Fractions

Consider an irrational number $x \in (0, 1)$ expanded as a dyadic (binary) fraction:

$$x = \frac{x_1}{2} + \frac{x_2}{2^2} + \frac{x_3}{2^3} + \cdots = (x_1, x_2, x_3, \dots), \quad x_n \in \{0, 1\}.$$

(We restrict to irrational x for uniqueness of the expansion; the rationals form a countable, hence measure-zero, set.) Among the first n digits, let p denote the number of zeros and $q = n - p$ the number of ones.

Theorem 3.1 (Hausdorff, 1914). *The set of those $x \in (0, 1)$ for which $\lim_{n \rightarrow \infty} \frac{p}{n} = \frac{1}{2}$ has Lebesgue measure 1.*

Proof. Measure of digit-constrained sets. The set of x whose binary expansion begins with prescribed digits $a_1, a_2, \dots, a_n$ is the set of irrationals in the interval

$$\frac{a_1}{2} + \cdots + \frac{a_n}{2^n} < x < \frac{a_1}{2} + \cdots + \frac{a_n + 1}{2^n},$$

which has Lebesgue measure $1/2^n$. The set of x for which exactly p of the first n digits

are zero is composed of $\binom{n}{p}$ such disjoint intervals, hence has measure

$$\binom{n}{p} \frac{1}{2^n}.$$

For $\varepsilon > 0$, let $M(n, \varepsilon)$ be the set of x for which $\left| \frac{p}{n} - \frac{1}{2} \right| \geq \varepsilon$. Its measure is

$$m(n, \varepsilon) = \sum_{\substack{p=0 \\ |p/n - 1/2| \geq \varepsilon}}^n \binom{n}{p} \frac{1}{2^n}.$$

Starting from the binomial identity $\sum_{p=0}^n \binom{n}{p} x^p y^q = (x+y)^n$ with $q = n-p$, Hausdorff applies the differential operator $x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}$ repeatedly. Setting $v = x - y$, $u = x + y$, one computes:

$$\begin{aligned} \sum_{p=0}^n \binom{n}{p} x^p y^q &= u^n, \\ \sum_{p=0}^n \binom{n}{p} (p-q) x^p y^q &= n u^{n-1} v, \\ \sum_{p=0}^n \binom{n}{p} (p-q)^2 x^p y^q &= n u^n + n(n-1) u^{n-2} v^2, \\ \sum_{p=0}^n \binom{n}{p} (p-q)^4 x^p y^q &= n(3n-2) u^n + n(n-1)(6n-8) u^{n-2} v^2 \\ &\quad + n(n-1)(n-2)(n-3) u^{n-4} v^4. \end{aligned}$$

Setting $x = y = \frac{1}{2}$ (so $u = 1$, $v = 0$), the fourth-power identity yields:

$$\sum_{p=0}^n \binom{n}{p} \frac{1}{2^n} (p-q)^4 = 3n^2 - 2n < 3n^2.$$

Rewriting $p - q = 2p - n$ and dividing by n^4 :

$$\sum_{p=0}^n \binom{n}{p} \frac{1}{2^n} \left(\frac{p}{n} - \frac{1}{2} \right)^4 < \frac{3}{16n^2}.$$

Bounding $m(n, \varepsilon)$. For $|p/n - 1/2| \geq \varepsilon$, we have $(p/n - 1/2)^4 \geq \varepsilon^4$, so:

$$\varepsilon^4 \cdot m(n, \varepsilon) \leq \sum_{\substack{p \\ |p/n - 1/2| \geq \varepsilon}} \binom{n}{p} \frac{1}{2^n} \left(\frac{p}{n} - \frac{1}{2} \right)^4 < \frac{3}{16n^2},$$

hence

$$m(n, \varepsilon) < \frac{3}{16\varepsilon^4 n^2}. \tag{2}$$

Let $M(\varepsilon) = \limsup_{n \rightarrow \infty} M(n, \varepsilon)$ be the set of x for which $|p/n - 1/2| \geq \varepsilon$ for infinitely many n . Define $S(n, \varepsilon) = \bigcup_{k=n}^{\infty} M(k, \varepsilon)$. Then $M(\varepsilon) = \bigcap_{n=1}^{\infty} S(n, \varepsilon)$ and

$$\lambda(S(n, \varepsilon)) \leq \sum_{k=n}^{\infty} m(k, \varepsilon) \leq \frac{3}{16\varepsilon^4} \sum_{k=n}^{\infty} \frac{1}{k^2}.$$

Since $\sum 1/k^2$ converges, $\lambda(S(n, \varepsilon)) \rightarrow 0$ as $n \rightarrow \infty$, so $\lambda(M(\varepsilon)) = 0$.

The set of x for which $\lim p/n \neq 1/2$ is $M = \bigcup_{k=1}^{\infty} M(1/k)$, a countable union of null sets, hence a null set. The complement of M has measure 1. $\square$

Note: Restriction to $[0, 1)$ and Why Not $\mathbb{R}$ Directly

Hausdorff's proof like Borel's works on the unit interval $[0, 1)$. We prove for $[0, 1)$ because its Lebesgue measure is 1, which lets us interpret measures as probabilities and use combinatorial counting (e.g. each n -digit binary string corresponds to an interval of length $1/2^n$).

The entire real line $\mathbb{R}$ has infinite Lebesgue measure, making a direct probabilistic argument impossible. To extend to $\mathbb{R}$, we use the measure-theoretic extension.

4 Measure-theoretic extension to the Real Line

Having established the result on $[0, 1)$ (whether by Borel's proof or Hausdorff's), we now extend it to all of $\mathbb{R}$.

Theorem 4.1. *Almost all real numbers in $\mathbb{R}$ are absolutely normal.*

Proof. Let $N^c \subset [0, 1)$ be the set of real numbers in the unit interval that are *not* absolutely normal. By Theorem 2.1, $\mu(N^c) = 0$.

Any $x \in \mathbb{R}$ can be written as $x = m + y$ where $m = \lfloor x \rfloor \in \mathbb{Z}$ and $y = x - \lfloor x \rfloor \in [0, 1)$. Since normality depends only on asymptotic digit frequencies, the finite prefix contributed by m has no effect. Hence x is absolutely normal if and only if its fractional part y is.

Write $\mathbb{R} = \bigcup_{m=-\infty}^{\infty} [m, m+1)$. For each $m \in \mathbb{Z}$, let $E_m \subset [m, m+1)$ be the set of non-normal numbers. Then $E_m = N^c + m$ and by translation invariance:

$$\mu(E_m) = \mu(N^c) = 0.$$

The set of all non-normal reals is $E = \bigcup_{m=-\infty}^{\infty} E_m$. By countable subadditivity:

$$\mu(E) \leq \sum_{m=-\infty}^{\infty} \mu(E_m) = 0.$$

Hence $\mathbb{R} \setminus E$ has full measure: almost all real numbers are absolutely normal. $\square$

Note: Why Restrict the Initial Proof to $[0, 1)$ Instead of $\mathbb{R}$?

To use geometric probability (or any probability measure), the total measure of the sample space must be exactly 1. The unit interval $[0, 1)$ satisfies this, allowing Lebesgue measure to serve directly as a probability distribution (e.g. each n -digit base- q string corresponds to an interval of measure $1/q^n$).

The real line $\mathbb{R}$ has *infinite* Lebesgue measure. One *cannot* define a uniform probability distribution on a space of infinite measure. Therefore, the probabilistic core of the argument must first be carried out on a finite measure space like $[0, 1)$, after which purely measure-theoretic properties—translation invariance and countable subadditivity—extend the result across all of $\mathbb{R}$.

5 Generalisation of Hausdorff's Proof to Arbitrary Bases

Hausdorff proved the result only for base 2. Following Weitz [4], we now carry out the generalisation.

5.1 Preliminary Lemmas on Null Sets

Lemma 5.1. *Every countable set is a null set.*

Proof. Let A be a countable set and let $(a_n)_{n \in \mathbb{N}}$ be an enumeration of A . Let $\varepsilon > 0$ be arbitrary. For each $n \in \mathbb{N}$, let I_n be an interval of length $\varepsilon/2^{n+1}$ containing a_n . Then $A \subseteq \bigcup_{n=0}^{\infty} I_n$ and

$$\sum_{n=0}^{\infty} \lambda(I_n) = \sum_{n=0}^{\infty} \frac{\varepsilon}{2^{n+1}} = \varepsilon.$$

Since ε was arbitrary, $\lambda(A) = 0$. □

Lemma 5.2. *The countable union of null sets is again a null set.*

Proof. Let $A_1, A_2, \dots$ be null sets and let $\varepsilon > 0$. For each $k \in \mathbb{N}^+$, since A_k is a null set, there exist countably many intervals $\{I_{k,j}\}_{j=1}^{\infty}$ such that $A_k \subseteq \bigcup_{j=1}^{\infty} I_{k,j}$ and $\sum_{j=1}^{\infty} \lambda(I_{k,j}) \leq \varepsilon/2^k$. Then $\bigcup_{k=1}^{\infty} A_k \subseteq \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{\infty} I_{k,j}$, a countable union of intervals, and

$$\sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \lambda(I_{k,j}) \leq \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon.$$

Since ε was arbitrary, $\lambda\left(\bigcup_k A_k\right) = 0$. □

5.2 Normality and Words

We define a finite sequence $w = b_1 b_2 \cdots b_n$ of digits in base r as an r -word and write $|w|$ for its length n . For $\alpha[n_1, n_2]_r := \alpha_{n_1, r} \cdots \alpha_{n_2, r}$ and an arbitrary word w , define

$$p_{w, n, \alpha, r} = \left| \{j \in \mathbb{N}^+ : j + |w| - 1 \leq n \wedge \alpha_{[j, j+|w|-1], r} = w\} \right|,$$

which counts overlapping occurrences of w among the first n digits of α .

Lemma 5.3. *If $r^m \alpha$ is simply normal in base r^n for all $n \in \mathbb{N}^+$ and all $m < n$, then α is normal in base r .*

Proof. Fix n and use r^n as the base. For $m \geq n$, write $m = kn + c$ with $0 \leq c < n$. Then $r^m \alpha = r^c \cdot (r^n)^k \alpha$. The digits of $(r^n)^k \alpha$ after the radix point coincide with those of α starting at position k . Multiplication by r^c shifts the grouping of base- r digits into base- r^n blocks by c positions. Hence the base- r^n digits of $r^m \alpha$ after the radix point agree, from some position onward, with those of $r^c \alpha$. Since $r^c \alpha$ is simply normal in base r^n by hypothesis (as $c < n$), so is $r^m \alpha$. $\square$

Lemma 5.4. *If α is normal in base r , then for all r -words w :*

$$\lim_{n \rightarrow \infty} \frac{p_{w, n, \alpha, r}}{n} = \frac{1}{r^{|w|}}.$$

Proof. Let $w = b_1 b_2 \cdots b_L$ be an r -word of length L . The word w corresponds to a single digit d_w in base $R = r^L$, namely $d_w = b_1 r^{L-1} + b_2 r^{L-2} + \cdots + b_L$.

Since α is normal in base r , by definition $r^m \alpha$ is simply normal in base r^L for every $m \in \mathbb{N}$. In particular, for $m = 0$, the digit d_w has limiting frequency $1/R = 1/r^L$ among the base- R digits of α . But the base- R digits of α are obtained by grouping consecutive blocks of L base- r digits starting at position 1.

This grouping captures the occurrences of w that begin at positions $1, L+1, 2L+1, \dots$. Now consider $r^c \alpha$ for $c = 1, 2, \dots, L-1$. The base- R digits of $r^c \alpha$ capture occurrences of w beginning at positions $c+1, c+L+1, c+2L+1, \dots$. Since α is normal, each $r^c \alpha$ is simply normal in base R , so each shifted grouping also yields frequency $1/r^L$ for the digit d_w .

The total count $p_{w, n, \alpha, r}$ of occurrences of w among the first n base- r digits is the sum of occurrences found in each of the L shifted groupings. More precisely, for $n = kL$, each grouping examines approximately $k = n/L$ base- R digits. Summing over $c = 0, 1, \dots, L-1$:

$$p_{w, n, \alpha, r} = \sum_{c=0}^{L-1} p_{d_w, \lfloor (n-c)/L \rfloor, r^c \alpha, R} \approx L \cdot \frac{n/L}{r^L} = \frac{n}{r^L}.$$

Dividing by n and taking $n \rightarrow \infty$ gives $\lim_{n \rightarrow \infty} p_{w, n, \alpha, r}/n = 1/r^L = 1/r^{|w|}$. $\square$

5.3 The Main Lemma

The following lemma generalises Hausdorff's fourth-moment computation from base 2 to arbitrary base r .

Lemma 5.5 (Main Lemma, generalising Hausdorff). *If $r \geq 2$ is an integer, then there exists a positive constant D such that for all $n \in \mathbb{N}^+$:*

$$\sum_{p=0}^n \binom{n}{p} \frac{(r-1)^{n-p}}{r^n} \left(\frac{p}{n} - \frac{1}{r} \right)^4 \leq \frac{D}{n^2}.$$

Proof. Fix positive integers s and n . Define functions recursively:

$$f_0(x, y) = \sum_{p=0}^n \binom{n}{p} x^{sp} y^{n-p}, \quad f_{k+1}(x, y) = x \frac{\partial}{\partial x} f_k(x, y) - y \frac{\partial}{\partial y} f_k(x, y).$$

Working with individual summands, one checks by induction that

$$f_k(x, y) = \sum_{p=0}^n \binom{n}{p} ((s+1)p - n)^k x^{sp} y^{n-p}. \quad (3)$$

By the binomial theorem, $f_0(x, y) = (x^s + y)^n$. Computing f_4 from this representation (a tedious but elementary exercise, nowadays easily done with a computer algebra system):

$$f_4(x, y) = n(x^s + y)^{n-4} Q,$$

where Q is a polynomial in x, y, s, n .

Now set $r = s + 1$, $x = 1/\sqrt[r]{r}$, and $y = (r - 1)/r$. Then $sx^s = s/r = (r - 1)/r = y$, so $sx^s - y = 0$, and several terms in Q vanish. The surviving terms simplify to:

$$f_4(x, y) = 3(r - 1)^2 n^2 + (r^3 - 7r^2 + 12r - 6) n.$$

This is a second-degree polynomial in n , so $f_4(x, y) \leq Cn^2$ for some constant C independent of n .

Substituting the expression (3) for $k = 4$:

$$\sum_{p=0}^n \binom{n}{p} \frac{(r-1)^{n-p}}{r^n} (rp - n)^4 \leq Cn^2.$$

Dividing both sides by $(rn)^4 = r^4 n^4$:

$$\sum_{p=0}^n \binom{n}{p} \frac{(r-1)^{n-p}}{r^n} \left(\frac{p}{n} - \frac{1}{r} \right)^4 \leq \frac{C}{r^4 n^2} =: \frac{D}{n^2}.$$

□

5.4 Almost All Numbers Are Normal

Theorem 5.6. *If $r \geq 2$ is an arbitrary base, then almost all numbers are simply normal in this base.*

Proof. We work in $[0, 1)$. Fix a base $r \geq 2$ and a digit $b \in \Sigma_r$. The set of numbers whose first n digits form a prescribed sequence is an interval of length $1/r^n$. The measure of the set of numbers with exactly p occurrences of b among their first n digits is

$$\binom{n}{p} \cdot (r-1)^{n-p} \cdot \frac{1}{r^n}. \quad (4)$$

For $\varepsilon > 0$, let $M_b(n, \varepsilon) = \{\alpha \in [0, 1) : |p_{b,n,\alpha,r}/n - 1/r| \geq \varepsilon\}$. Using (4):

$$\lambda(M_b(n, \varepsilon)) = \sum_{\substack{p=0 \\ |p/n - 1/r| \geq \varepsilon}}^n \binom{n}{p} \frac{(r-1)^{n-p}}{r^n}.$$

Since $|p/n - 1/r| \geq \varepsilon$ implies $(p/n - 1/r)^4 \geq \varepsilon^4$:

$$\varepsilon^4 \cdot \lambda(M_b(n, \varepsilon)) \leq \sum_{p=0}^n \binom{n}{p} \frac{(r-1)^{n-p}}{r^n} \left(\frac{p}{n} - \frac{1}{r}\right)^4 \leq \frac{D}{n^2},$$

by Theorem 5.5. Therefore:

$$\lambda(M_b(n, \varepsilon)) \leq \frac{D}{\varepsilon^4 n^2}. \quad (5)$$

Let $M_b(\varepsilon)$ be the set of $\alpha \in [0, 1)$ with $|p_{b,n,\alpha,r}/n - 1/r| \geq \varepsilon$ for infinitely many n . Define $S_b(m, \varepsilon) = \bigcup_{n=m}^{\infty} M_b(n, \varepsilon)$. Then $M_b(\varepsilon) = \bigcap_{m=1}^{\infty} S_b(m, \varepsilon)$ and

$$\lambda(S_b(m, \varepsilon)) \leq \sum_{n=m}^{\infty} \lambda(M_b(n, \varepsilon)) \leq \frac{D}{\varepsilon^4} \sum_{n=m}^{\infty} \frac{1}{n^2}.$$

Since $\sum 1/n^2 < \infty$, the tail can be made arbitrarily small. Hence $\lambda(M_b(\varepsilon)) = 0$ for every $\varepsilon > 0$.

The set $M_b = \bigcup_{k=1}^{\infty} M_b(1/k)$ of numbers violating (1) for digit b is a countable union of null sets, hence null by Theorem 5.2. The set of numbers not simply normal in base r is $\bigcup_{b=0}^{r-1} M_b$, a finite union of null sets, hence null. $\square$

Corollary 5.7. *If $r \geq 2$ is an arbitrary base, then almost all numbers are normal in this base.*

Proof. Multiplying each element of $[0, 1)$ by r^m for some $m > 0$ “spreads” the set over $[0, r^m)$. Since the set of numbers not simply normal in base r in each interval $[j, j+1)$ is a null set (by Theorem 5.6), their union is also null by Theorem 5.2.

Another application of Theorem 5.2 shows that the set of $\alpha \in [0, 1)$ such that $r^m \alpha$ is not simply normal in base r for at least one $m \in \mathbb{N}$ is a null set. The same argument applies with bases $r^2, r^3, \dots$. A final application of Theorem 5.2 (over the countable set of exponents) gives the result. $\square$

Corollary 5.8 (Borel’s Theorem). *Almost all numbers are absolutely normal.*

Proof. By Theorem 5.7, for each base $r \geq 2$, the set of numbers not normal in base r is a null set. Since $\{2, 3, 4, \dots\}$ is countable, the union of these null sets is null by Theorem 5.2. $\square$

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